MSG No. 197.8 $I231'$ [Type II, cubic]

Table 1: Wyckoff site: 2a, site symmetry: 23.1'

No.	position	mapping
1	$[0,\ 0,\ 0]$	[1,2,3,4,5,6,7,8,9,10,11,12,25,26,27,28,29,30,31,32,33,34,35,36]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[13,14,15,16,17,18,19,20,21,22,23,24,37,38,39,40,41,42,43,44,45,46,47,48]

Table 2: Wyckoff site: 6b, site symmetry: 222..1'

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	[1,2,3,4,25,26,27,28]
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[5,8,10,12,29,32,34,36]
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[6,7,9,11,30,31,33,35]
4	$\left[\frac{1}{2},\ 0,\ 0\right]$	[13,14,15,16,37,38,39,40]
5	$\left[0,\ \frac{1}{2},\ 0\right]$	[17,20,22,24,41,44,46,48]
6	$\left[0,\ 0,\ \frac{1}{2}\right]$	[18,19,21,23,42,43,45,47]

Table 3: Wyckoff site: 8c, site symmetry: .3.1'

No.	position	mapping
1	$[x,\ x,\ x]$	[1,5,6,25,29,30]
2	$[x,\ -x,\ -x]$	[2,10,11,26,34,35]
3	$[-x,\ x,\ -x]$	[3,7,12,27,31,36]
4	$[-x,\ -x,\ x]$	[4,8,9,28,32,33]
5	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	[13,17,18,37,41,42]
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	[14,22,23,38,46,47]
7	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	[15,19,24,39,43,48]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	[16,20,21,40,44,45]

Table 4: Wyckoff site: 12d, site symmetry: 2..1'

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,2,25,26]
2	$[-x,\ 0,\ 0]$	[3,4,27,28]
3	$[0,\ x,\ 0]$	[5,12,29,36]
4	$[0,\ 0,\ x]$	[6,9,30,33]
5	$[0,\ 0,\ -x]$	[7,11,31,35]
6	$[0,\ -x,\ 0]$	[8,10,32,34]
7	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[13,14,37,38]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[15,16,39,40]
9	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[17,24,41,48]
10	$\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	[18,21,42,45]
11	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	[19,23,43,47]
12	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[20,22,44,46]

Table 5: Wyckoff site: `12e`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,2,25,26]`
2	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[3,4,27,28]`
3	$\left[0,\ x,\ \frac{1}{2}\right]$	`[5,12,29,36]`
4	$\left[\frac{1}{2},\ 0,\ x\right]$	`[6,9,30,33]`
5	$\left[\frac{1}{2},\ 0,\ -x\right]$	`[7,11,31,35]`
6	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[8,10,32,34]`
7	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[13,14,37,38]`
8	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	`[15,16,39,40]`
9	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[17,24,41,48]`
10	$\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[18,21,42,45]`
11	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[19,23,43,47]`
12	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[20,22,44,46]`

Table 6: Wyckoff site: `24f`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,25]`
2	$[x,\ -y,\ -z]$	`[2,26]`
3	$[-x,\ y,\ -z]$	`[3,27]`
4	$[-x,\ -y,\ z]$	`[4,28]`
5	$[z,\ x,\ y]$	`[5,29]`
6	$[y,\ z,\ x]$	`[6,30]`
7	$[-y,\ z,\ -x]$	`[7,31]`
8	$[-z,\ -x,\ y]$	`[8,32]`
9	$[-y,\ -z,\ x]$	`[9,33]`
10	$[z,\ -x,\ -y]$	`[10,34]`
11	$[y,\ -z,\ -x]$	`[11,35]`
12	$[-z,\ x,\ -y]$	`[12,36]`
13	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[13,37]`
14	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[14,38]`
15	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[15,39]`
16	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[16,40]`
17	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$	`[17,41]`
18	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[18,42]`
19	$\left[\frac{1}{2}-y,\ z+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[19,43]`
20	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ y+\frac{1}{2}\right]$	`[20,44]`
21	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ x+\frac{1}{2}\right]$	`[21,45]`
22	$\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	`[22,46]`
23	$\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	`[23,47]`
24	$\left[\frac{1}{2}-z,\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[24,48]`